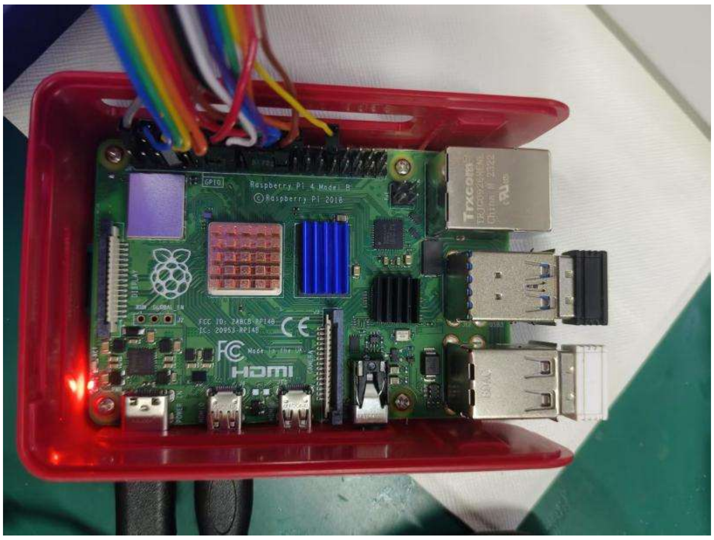
Driven LED Screen Based on Raspberry Pie

Overview:

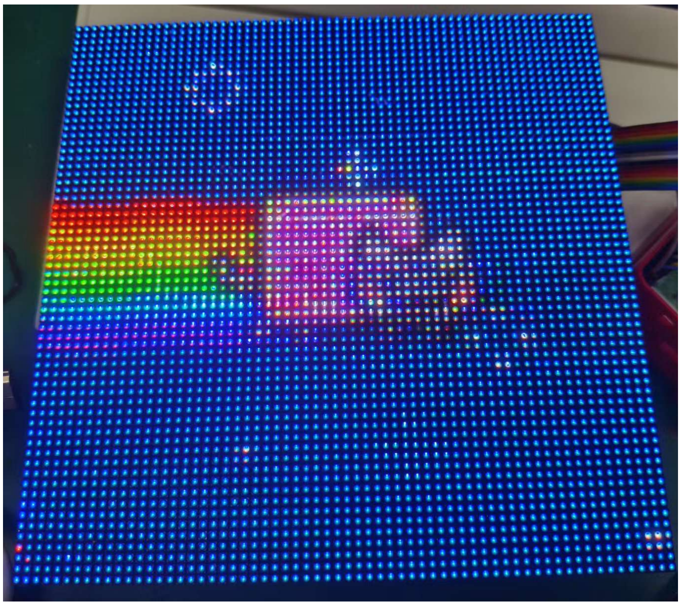
This technical document provides a comprehensive and in-depth description of how to start from scratch, combine a raspberry pie with an LED screen, successfully light up the LED screen drive, and conduct a series of creative and fun experiments.

一、Device Preparation

1、Raspberry Pie 4B



2、64*64 LED Screen



3、HDMI External Display
4、Mouse & Keyboard

二、Preparations

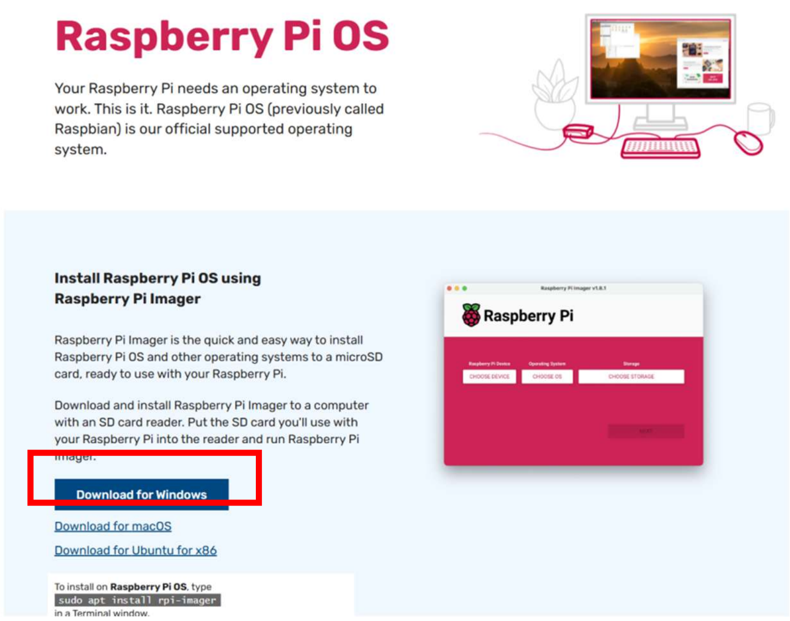
1、Hardware wiring

Connection	Pin	Pin	Connection
-	1	2	-
[3] G1	3	4	-
[3] B1	5	6	GND
LAT/STB	7	8	[3] R1
-	9	10	E
CLK	11	12	OE-
[1] G1	13	14	-
A	15	16	B
-	17	18	C
[1] B2	19	20	-
[1] G2	21	22	D
[1] R1	23	24	[1] R2
-	25	26	[1] B1
-	27	28	-
[2] G1	29	30	-
[2] B1	31	32	[2] R1
[2] G2	33	34	-
[2] R2	35	36	[3] G2
[3] R2	37	38	[2] B2
-	39	40	[3] B2

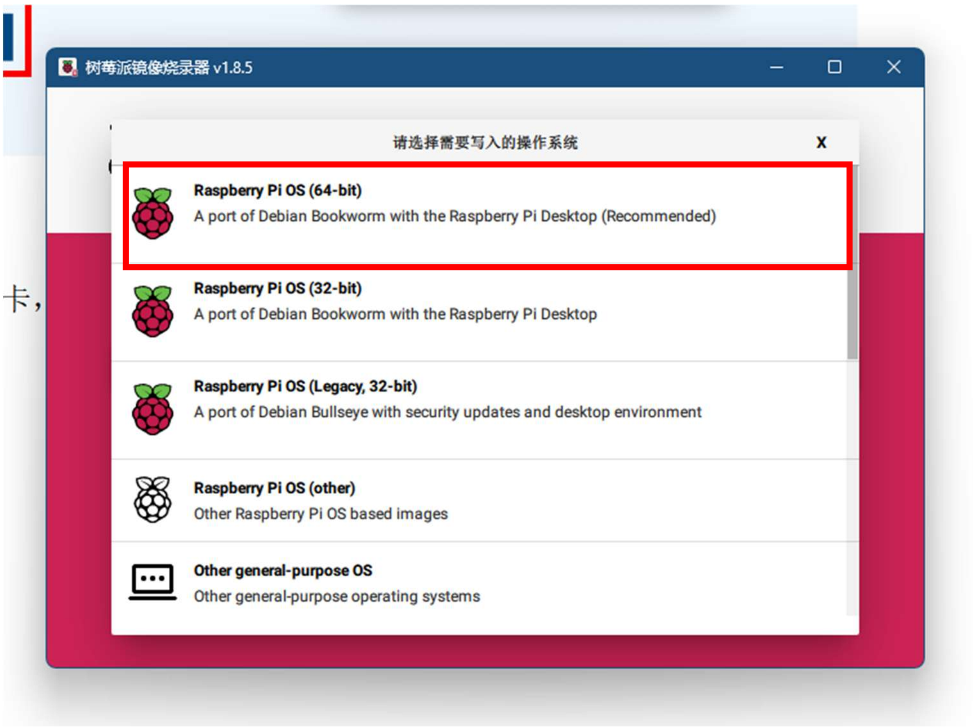
Just Connect Lines with Smiley Faces！

2、Raspberry Pi System Burning

Enter：https://www.raspberrypi.com/software/



After the download and installation are completed, insert the memory card, select the 64-bit system, and click "Burn".



3、Power on and start up, Configure the username and password (Current password: 123456), connect WIFI。

4、Connect the display screen using an HDMI cable

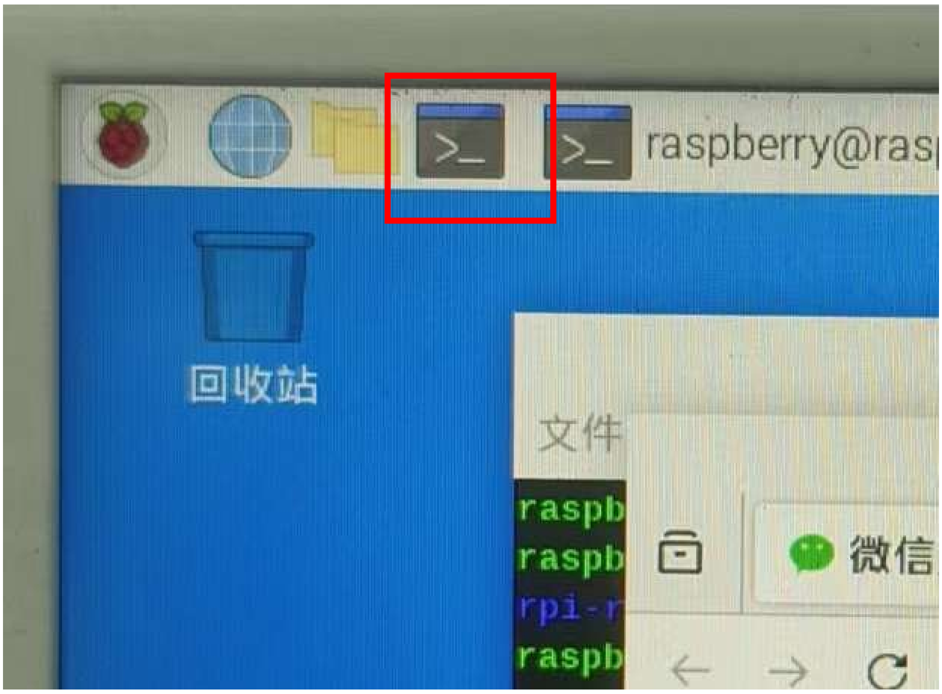
三、Specific Steps

1、Open the command line, create a new folder, then change the directory to that folder, and enter the command

"*git clone https://github.com/hzeller/rpi-rgb-led-matrix.git*"

to download the project to your local machine.

If it is a compressed file, you will need to extract it. (*Current project directory: ./home/project/rpi-rgb-led-matrix*)



2、Disable the onboard audio driver of the Raspberry Pi and modify the cmdline.txt file

（1）Disable the driver

Open the command line and enter：

cat <<EOF | sudo tee /etc/modprobe.d/blacklist-rgb-matrix.conf

blacklist snd_bcm2835

EOF

Then input *sudo update-initramfs -u* upload initramfs

Finally input *sudo reboot* to restart

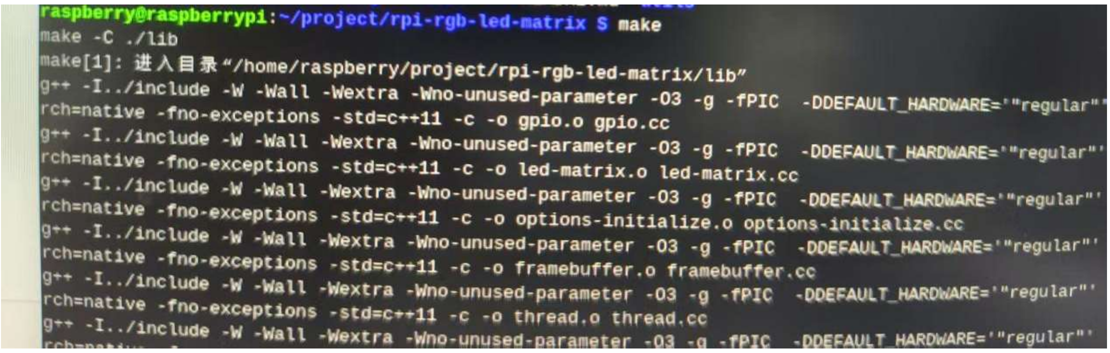
(2) Modify the file

Enter: *sudo nano /boot/cmdline.txt*

After entering the file, add "*isolcpus=3*" at the end of the line followed by a space.

Input: *CTRL+O*, press Enter, then *CTRL+X*.

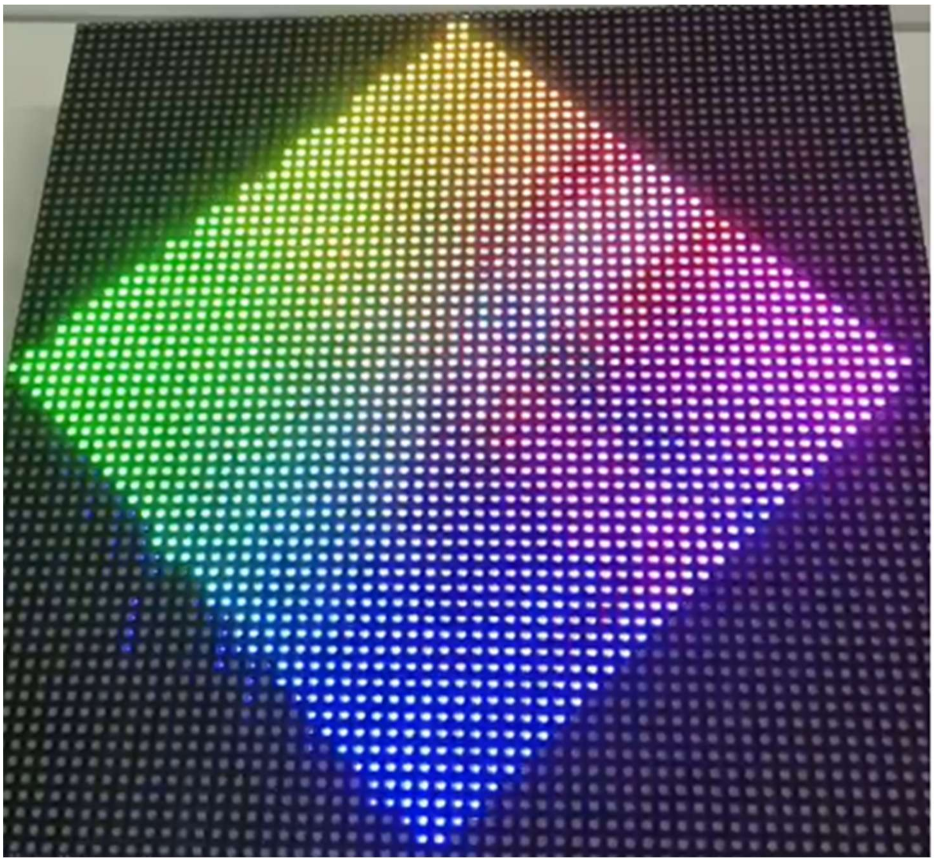
3、Enter the directory "*/home/project/rpi-rgb-led-matrix*", then type "*make*" to compile the project.



5、After the compilation is completed, go to the directory:

/home/project/rpi-rgb-led-matrix/examples-api-use, and enter the command:

sudo ./demo -D0 --led-rows=64 --led-cols=64. The following effect will appear：



If an error message appears:

=== snd_bcm2835: found that the Pi sound module is loaded. ===

Don't use the built-in sound of the Pi together with this lib; it is known to be incompatible and cause trouble and hangs (you can still use external USB sound adapters).

See Troubleshooting section in README how to disable the sound module.

You can also run with --led-no-hardware-pulse to avoid the incompatibility,

but you will have more flicker.

Exiting; fix the above first or use --led-no-hardware-pulse

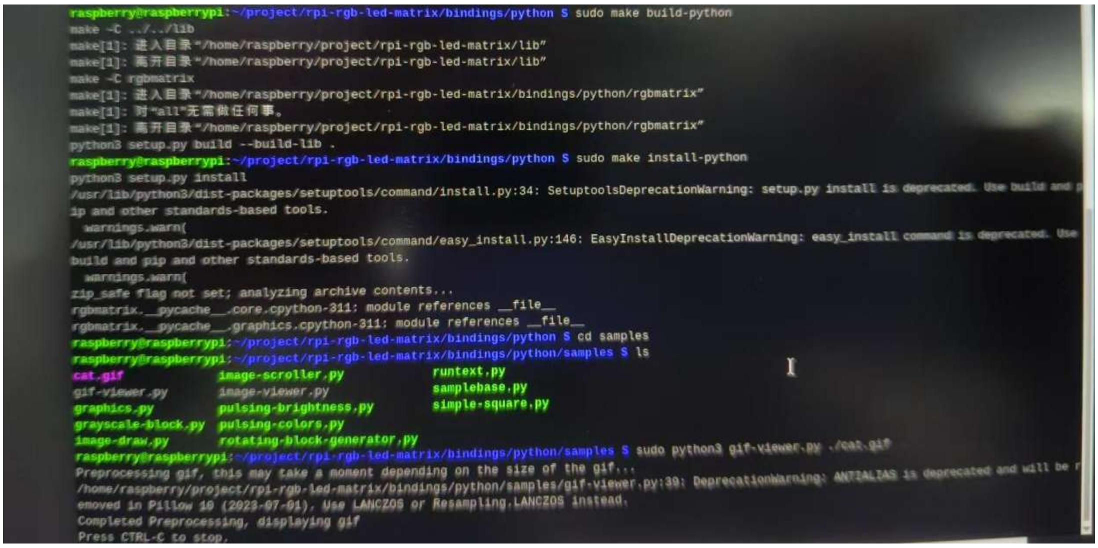
Then, the second step needs to be repeated: disable the onboard audio driver of the Raspberry Pi, and after restarting, input "lsmod | grep snd_bcm2835" to verify if the driver has been loaded. If it has not been loaded, it is OK.

6、Configuring the Python environment：

In the directory *./home/project/rpi-rgb-led-matrix/bindings/python*, enter the following commands: *sudo apt-get update && sudo apt-get install python3-dev cython3 -y*

Then enter： sudo make build-python

Finally enter： sudo make install-python



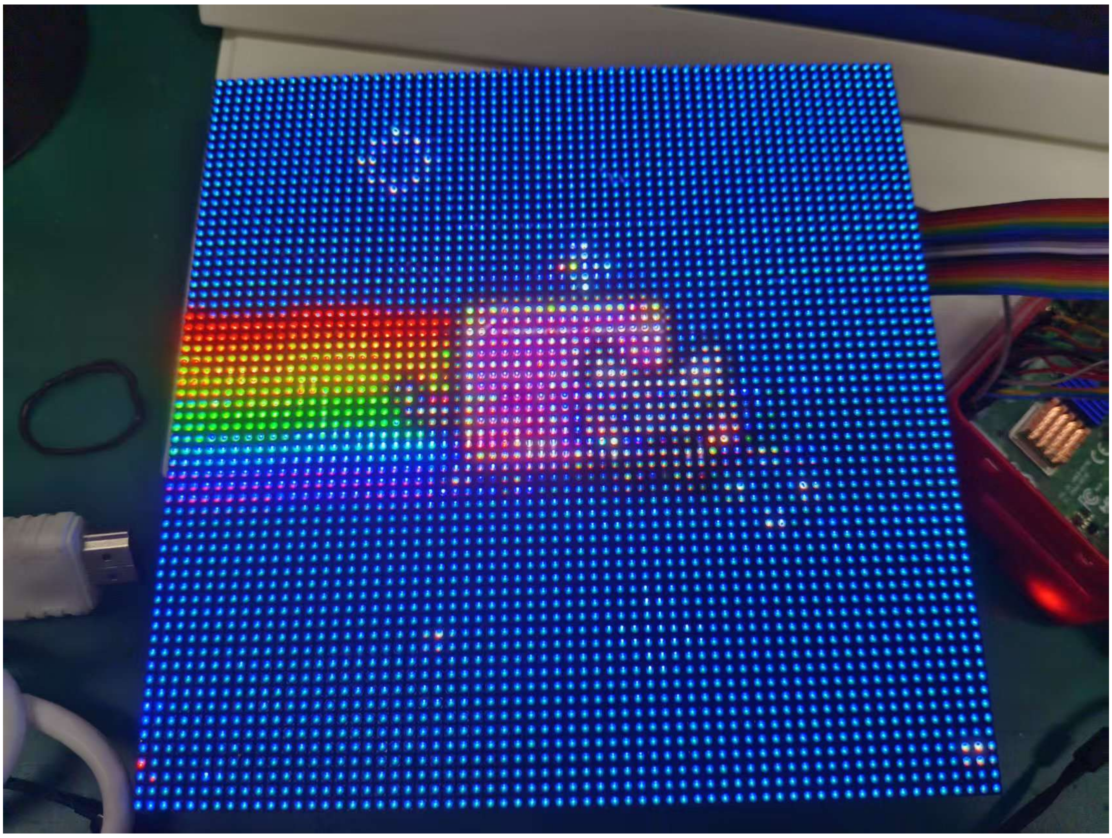
7、Enter the directory: */home/project/rpi-rgb-led-matrix/bindings/python/samples/ and modify the file gif-viewer.py*

Input: sudo nano gif-viewer.py Modify the screen resolution to 64*64

```
try:
    num_frames = gif.n_frames
except Exception:
    sys.exit("provided image is not a gif")


# Configuration for the matrix
options = RGBMatrixOptions()
options.rows = 32
options.cols = 32
options.chain_length = 1
options.parallel = 1
options.hardware_mapping = 'regular'  # If you have an Adafruit HAT: 'adafruit-hat'

matrix = RGBMatrix(options = options)

# Preprocess the gifs frames into canvases to improve playback performance
canvases = []
print("Preprocessing gif, this may take a moment depending on the size of the gif...")
for frame_index in range(0, num_frames):
    gif.seek(frame_index)
    # must copy the frame out of the gif, since thumbnail() modifies the image in-place
    frame = gif.copy()
    frame.thumbnail((matrix.width, matrix.height), Image.ANTIALIAS)
    canvas = matrix.CreateFrameCanvas()
    canvas.SetImage(frame.convert("RGB"))
    canvases.append(canvas)
# Close the gif file to save memory now that we have copied out all of the frames
```

8、Enter the command "*sudo python3 gif-viewer.py ./cat.gif*"(assuming there is a "cat.gif" file in the directory, which is currently included).

This will produce the following result：



To achieve the effects of other GIF animations, simply go online to download them.